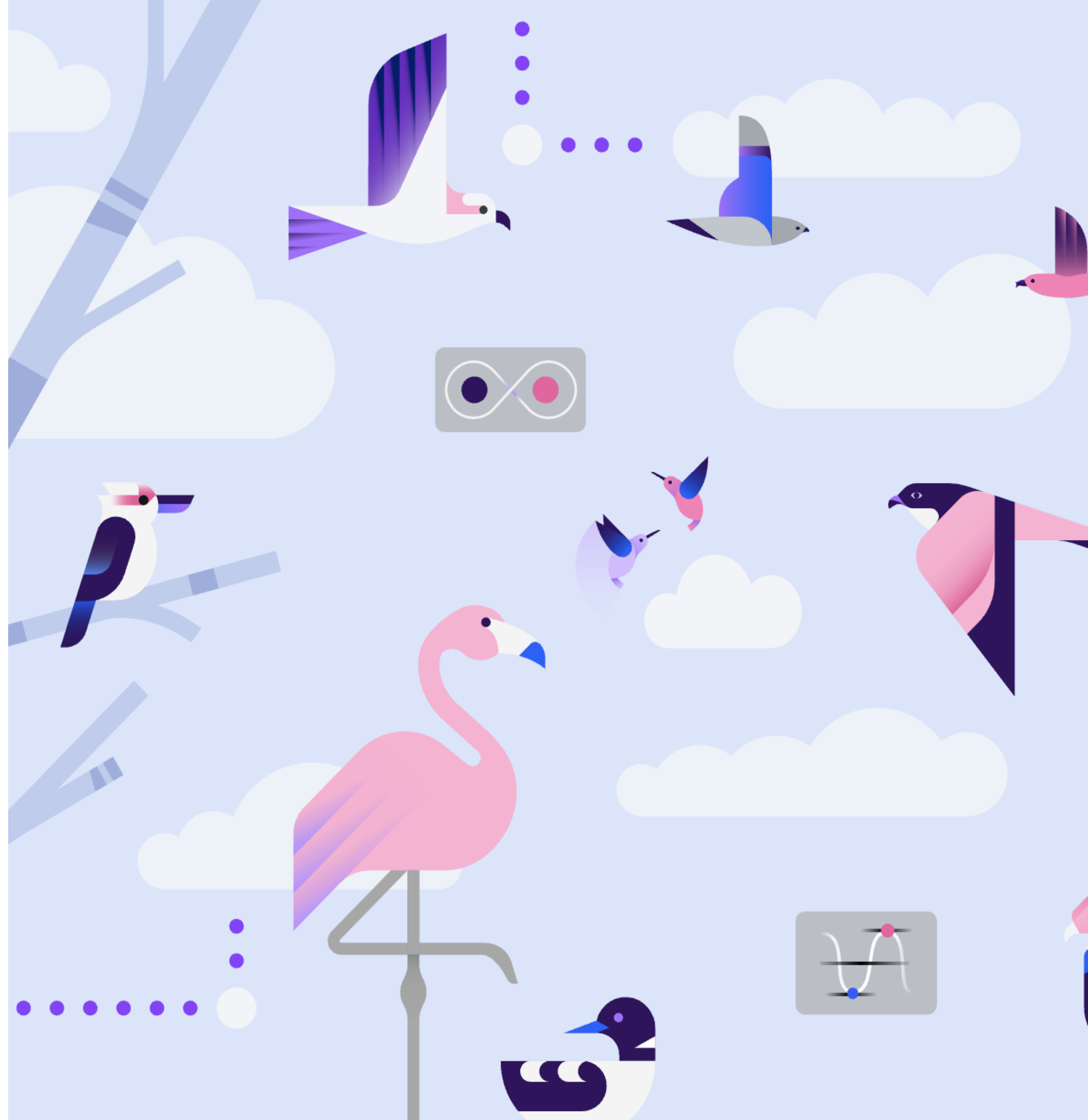


Introduction to IBM Quantum and Qiskit

Dr Sophie Engineer
Quantum Developer Advocate
IBM Quantum
Qiskit Fall Fest 2026

*University of Ibadan Quantum Computing Club,
Nigeria*



Agenda

Part 1:

- What problems can quantum computers solve?
- Intro to quantum computing

Part 2:

- Connecting to the IBM Quantum Platform
- Running jobs on the real hardware vs simulations



IBM Quantum mission

Bring useful
quantum
computing to
the world



The **new** wave of computing



Classical computer
Well suited for many problems



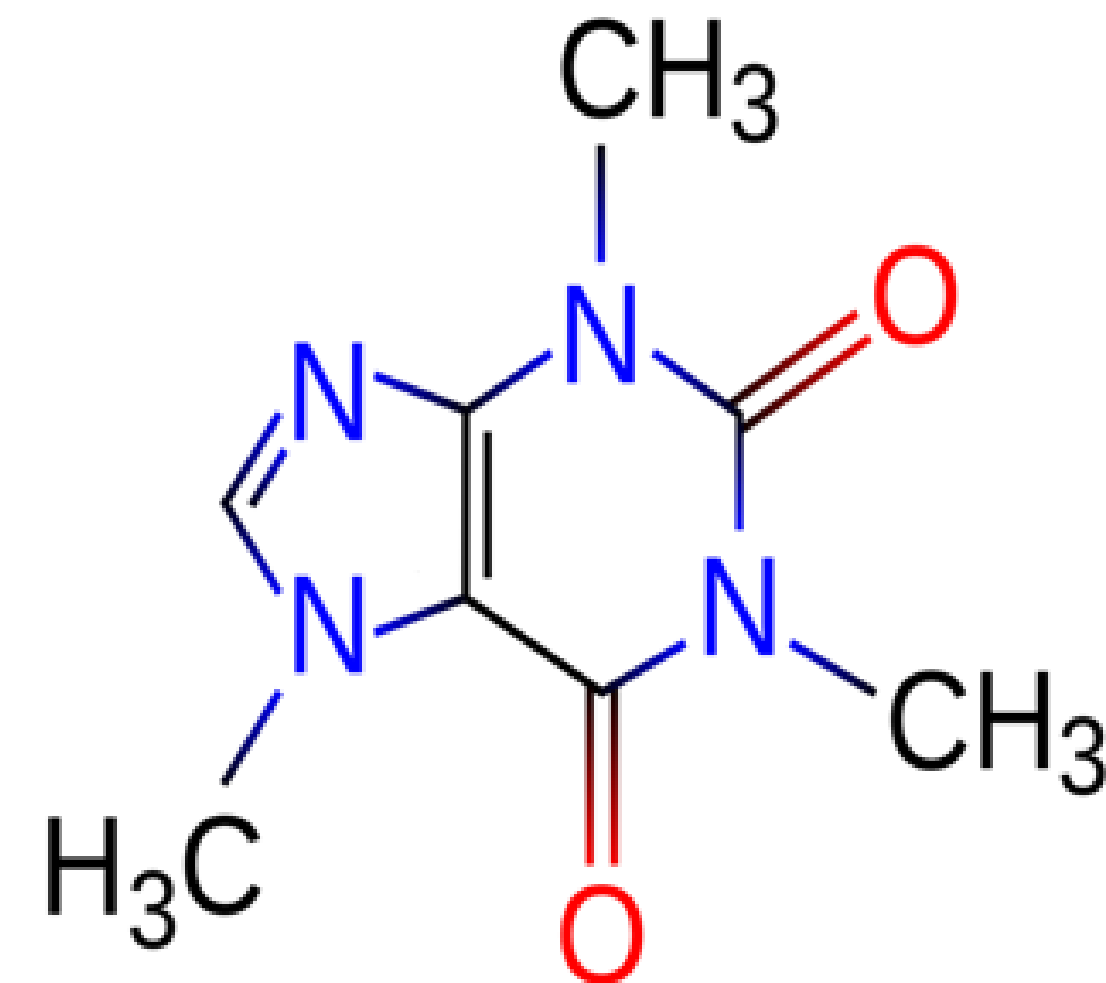
Quantum computer
Unlock classically intractable problems

Computing with caffeine

If our best classical computers are so powerful, shouldn't we be able to perfectly simulate molecules and chemical reactions?

This would allow us to accelerate discovery of new compounds and processes for healthcare, materials, alloys, and sustainable energy creation.

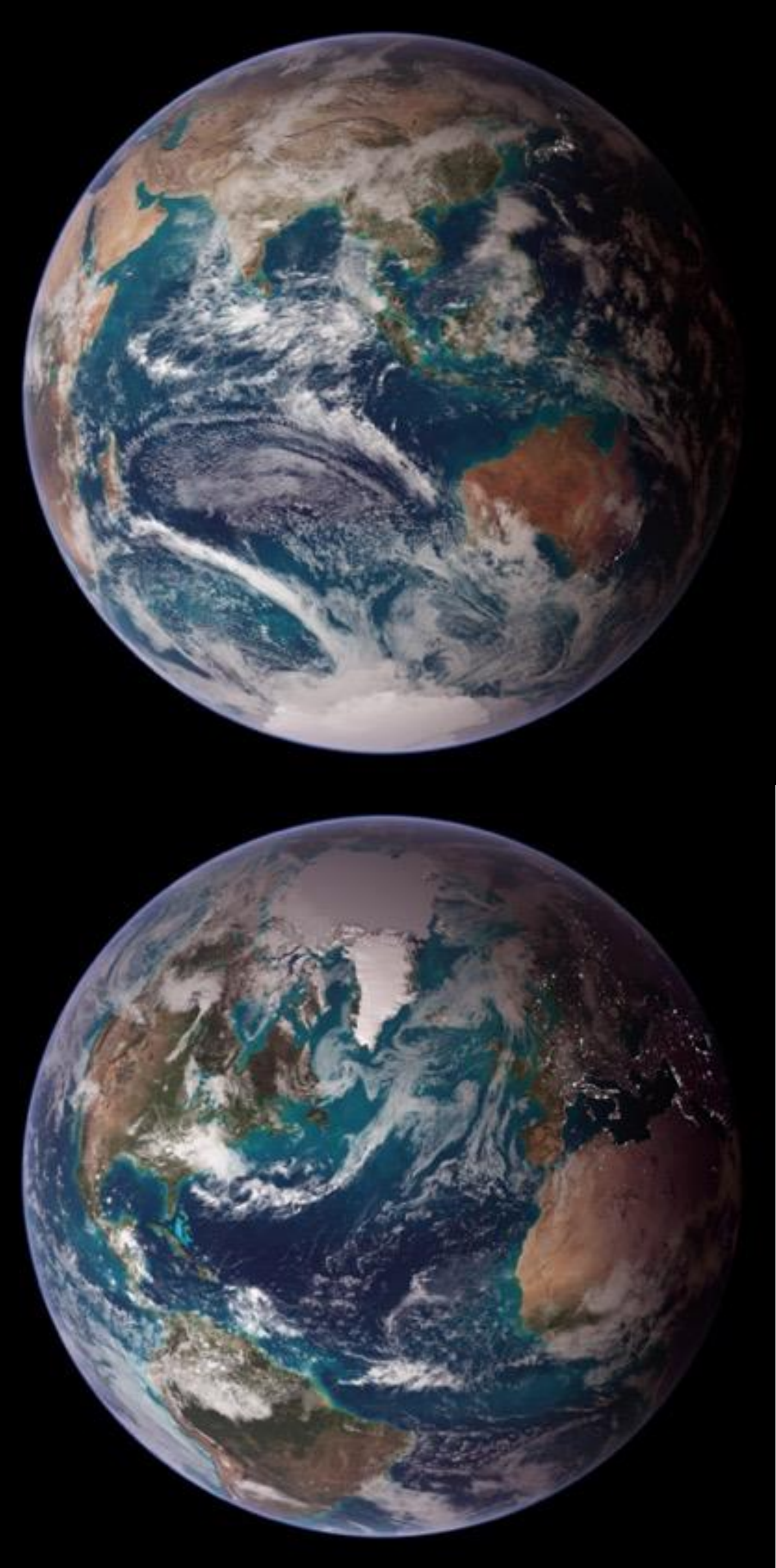
Let's consider caffeine ...



Computing with caffeine

We would need approximately 10^{48} bits to represent the energy configuration of a single caffeine molecule at a single instant in a classical computer.

This is 1 to 10% of the total number of atoms in the Earth.

$$10^{48} = \begin{array}{l} 1,000,000,000,000,000, \\ 000,000,000,000,000,000, \\ 000,000,000,000,000 \end{array}$$


Computing with caffeine

Although it's impossible to completely represent the molecular configuration of caffeine on today's most powerful super computers, we could represent it using 160 **logical qubits**.

IBM Quantum



What are these problems?

Modeling molecules, atoms, electrons, and quarks with **unprecedented accuracy**



Developing lighter, longer-lasting batteries for electric vehicles, electronics, and energy grid storage



Designing lighter, stronger materials to allow planes to be more efficient and to need less maintenance



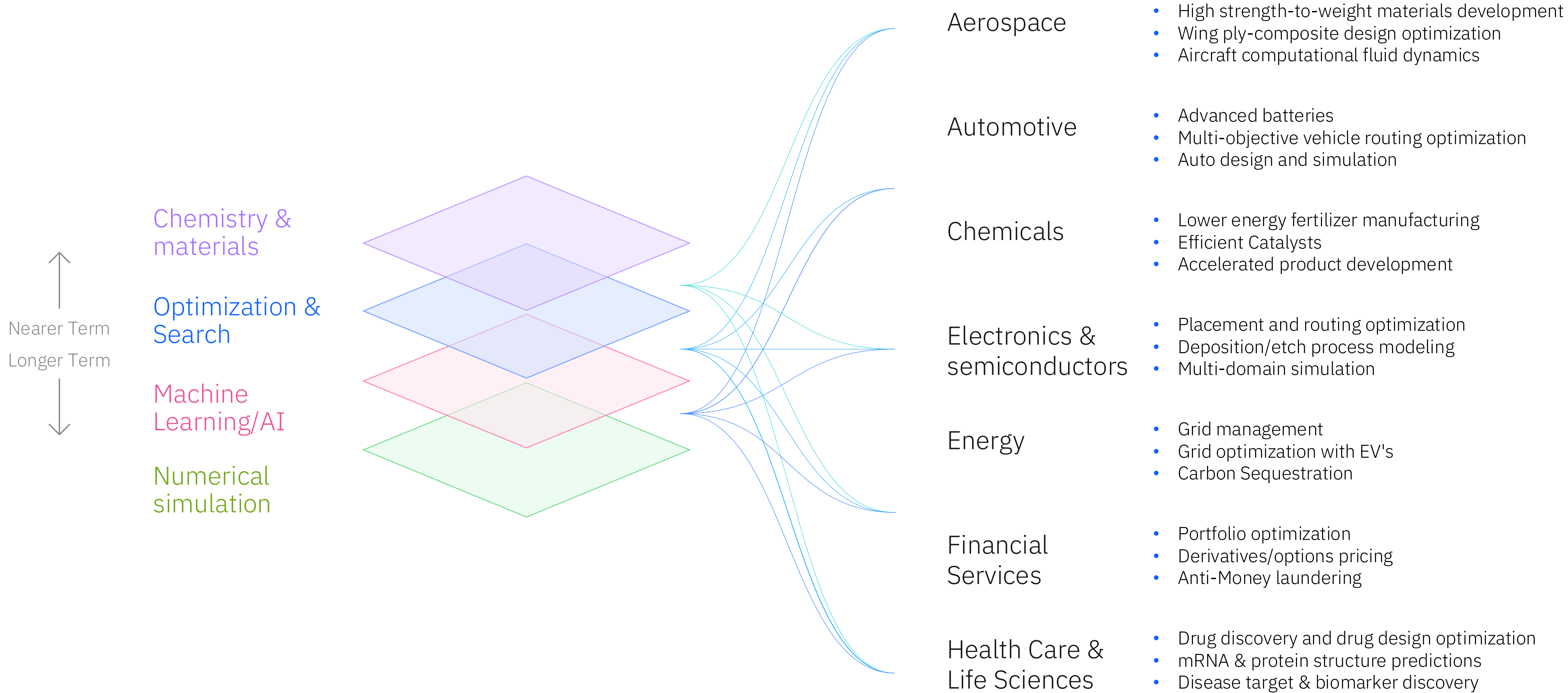
Discovering new classes of antibiotics to counter the emergence of multidrug-resistant bacterial strains



Designing optimal superconductors for MRI, electromobility, and renewable energies

Quantum computing is expected to have impact across industries

Example use cases



Bits and classical logic circuits

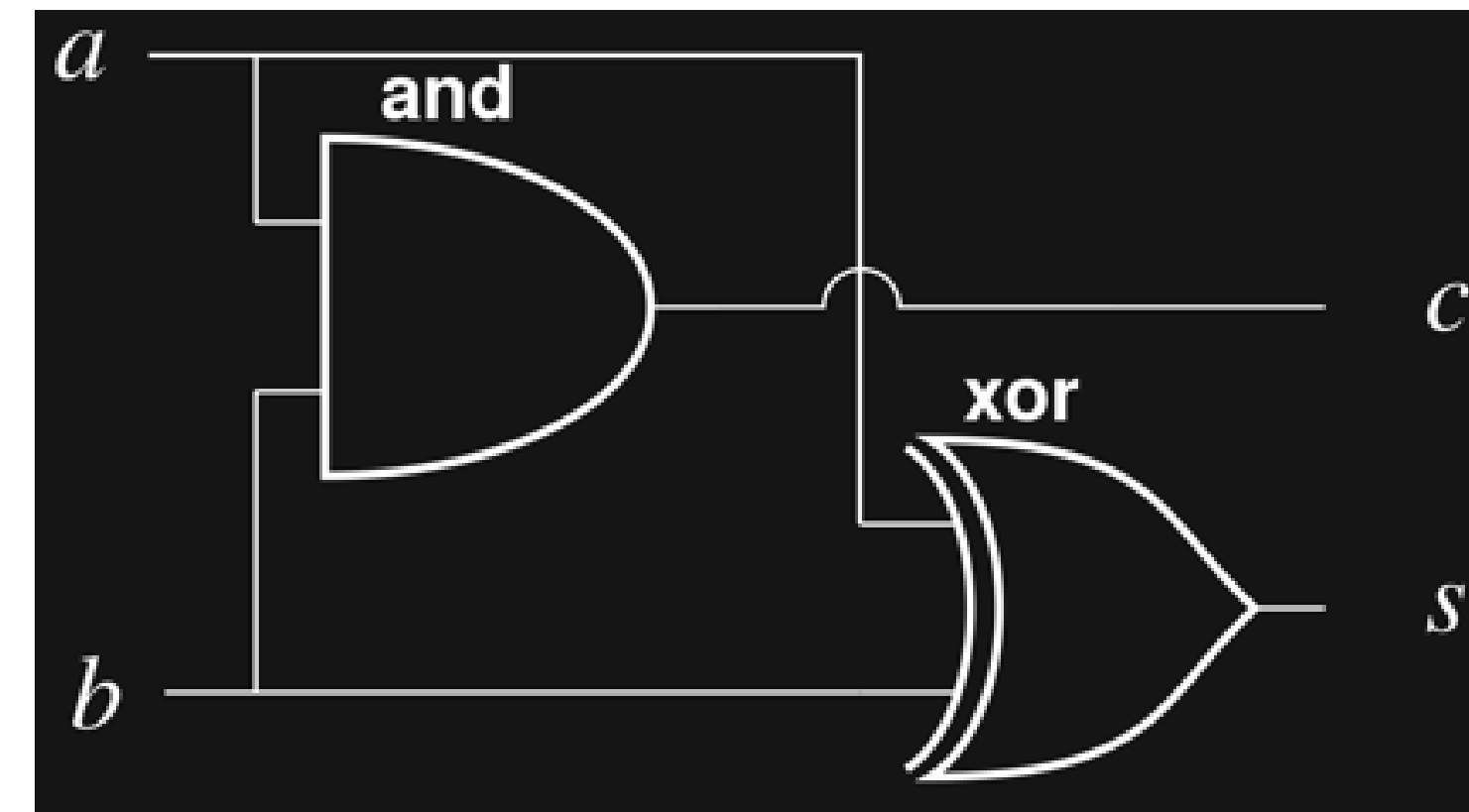
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•

•

1

A **bit** is a controllable classical object that is the unit of information



A **classical logic** circuit is a set of gate operations on bits and is the unit of computation

The limit of bits

For decades we've been simplifying nature into **1**s and **0**s because that was the only way we could **manage** to create a useful and scalable system of computation.

```
00100110111001001000100100100110010011100101110
01111100101001000111000100010010100010010010101
01001010101110010011011100100100010010010011001
00111001011100111110010100100011100010001001010
0010010010101010010101011101110011100101011110
```

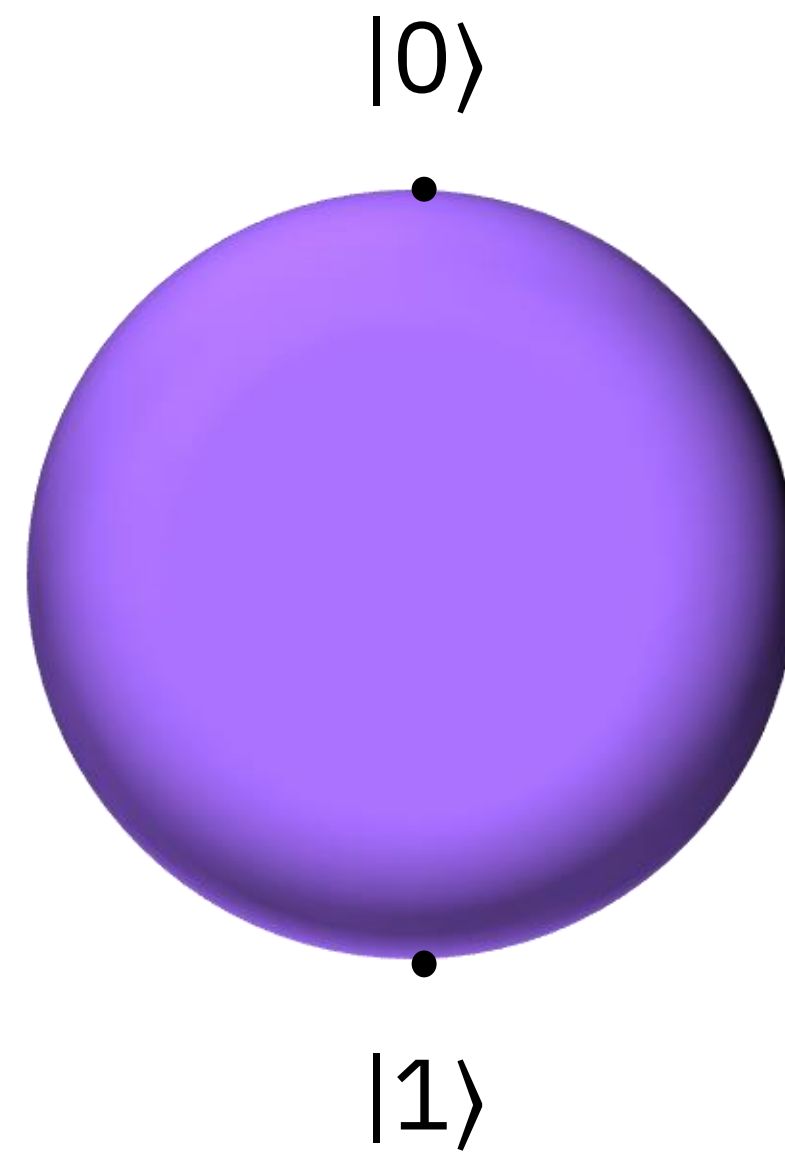
The limit of bits

For decades we've been simplifying nature into **1**s and **0**s because that was the only way we could **manage** to create a useful and scalable system of computation.

But the future isn't
just **1**s and **0**s.

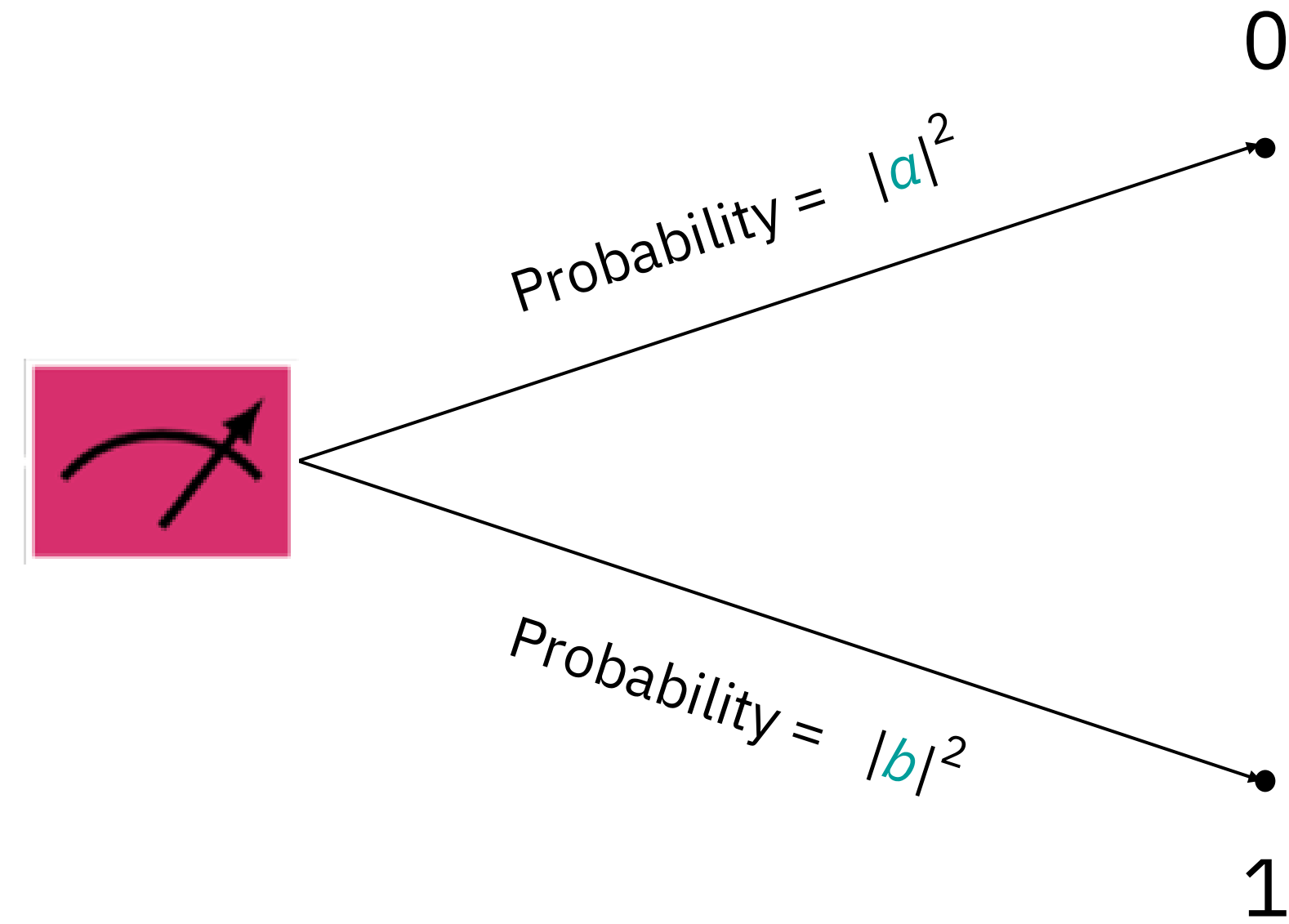
```
00100110111001001000100100100110010011100101110
01111100101001000111000100010010100010010010101
01001010101110010011011100100100010010010011001
00111001011100111110010100100011100010001001010
0010010010101010010101011101110011100101011110
```

Bits and qubits



A qubit's **state** is a combination of $|0\rangle$ and $|1\rangle$:
 $a |0\rangle + b |1\rangle$

This means that a single qubit contains
two pieces of information.



When we measure a qubit, it becomes
0 or **1** based on probability.

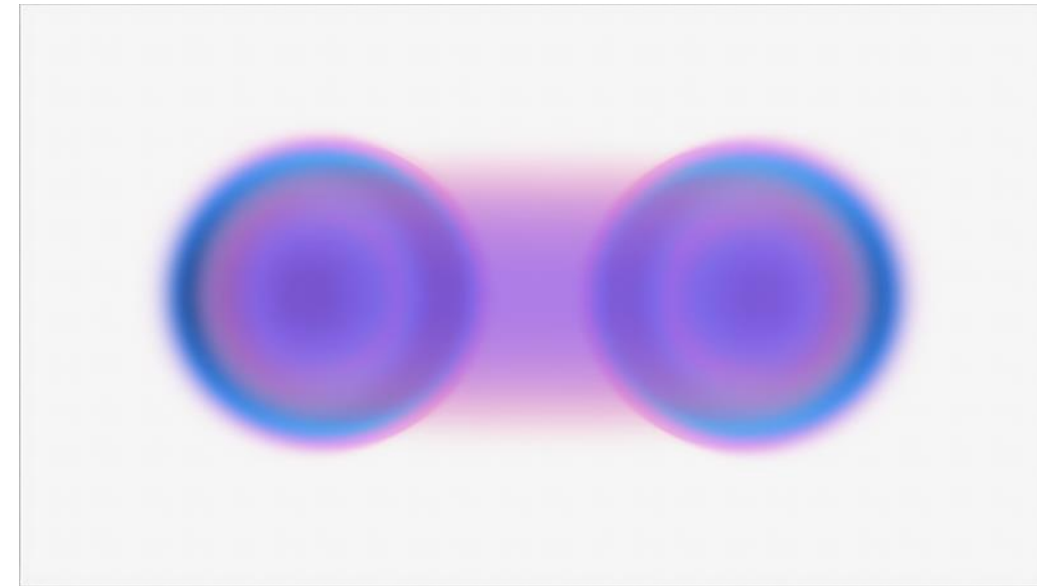
Uniquely quantum

Some problems are classically intractable and will never be solvable with traditional computers



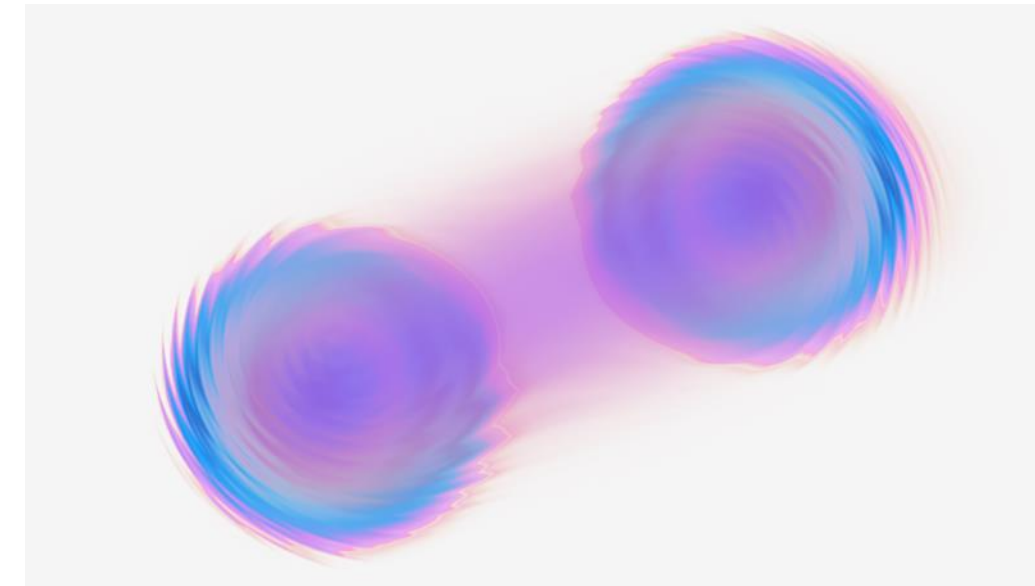
Superposition

A quantum system existing in a complex linear combination of two states until it is measured



Entanglement

Information shared jointly between entangled pairs or groups



Interference

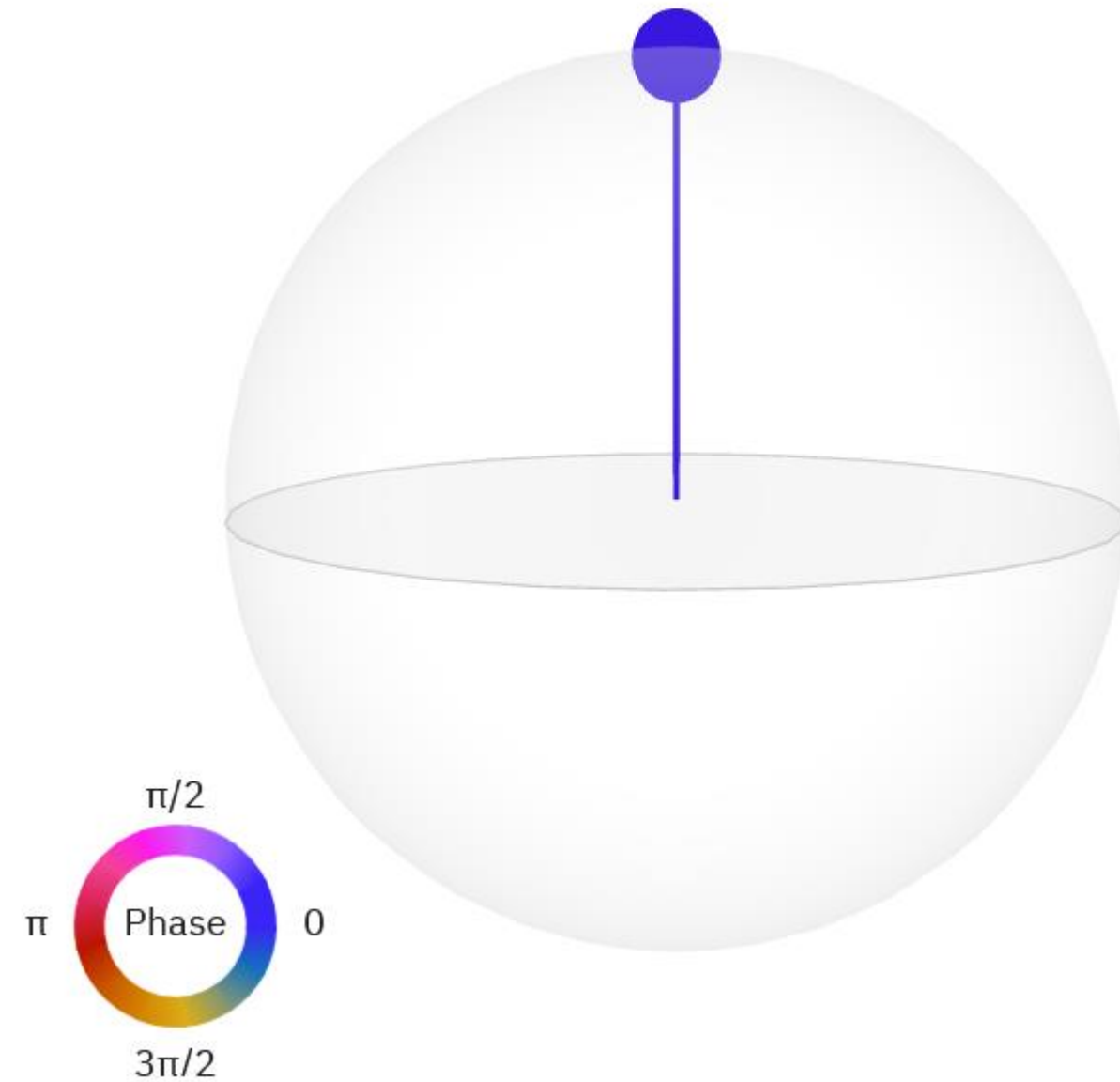
Interaction that affects likelihood of solutions

Moore's law: the number of transistors in a classical integrated circuit doubles about every two years ... but we are approaching the end due to physical limitations

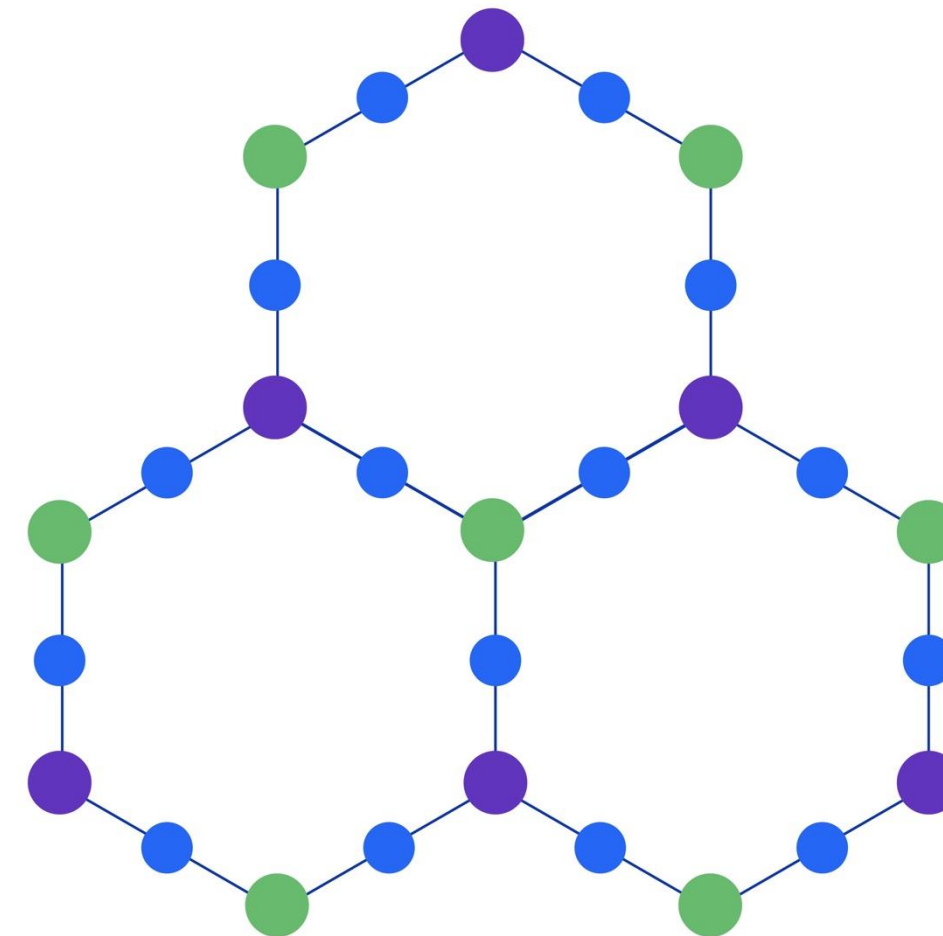
[Approaching the physical limit: IBM created the world's first 2 nm node chip in 2021, with transistors as small as 10 silicon atoms](#)

These Quantum concepts can reduce the number of computational steps required for certain algorithms & At a ~100 qubit scale, with sufficient circuit depth and complexity, classical computers can no longer simulate exactly

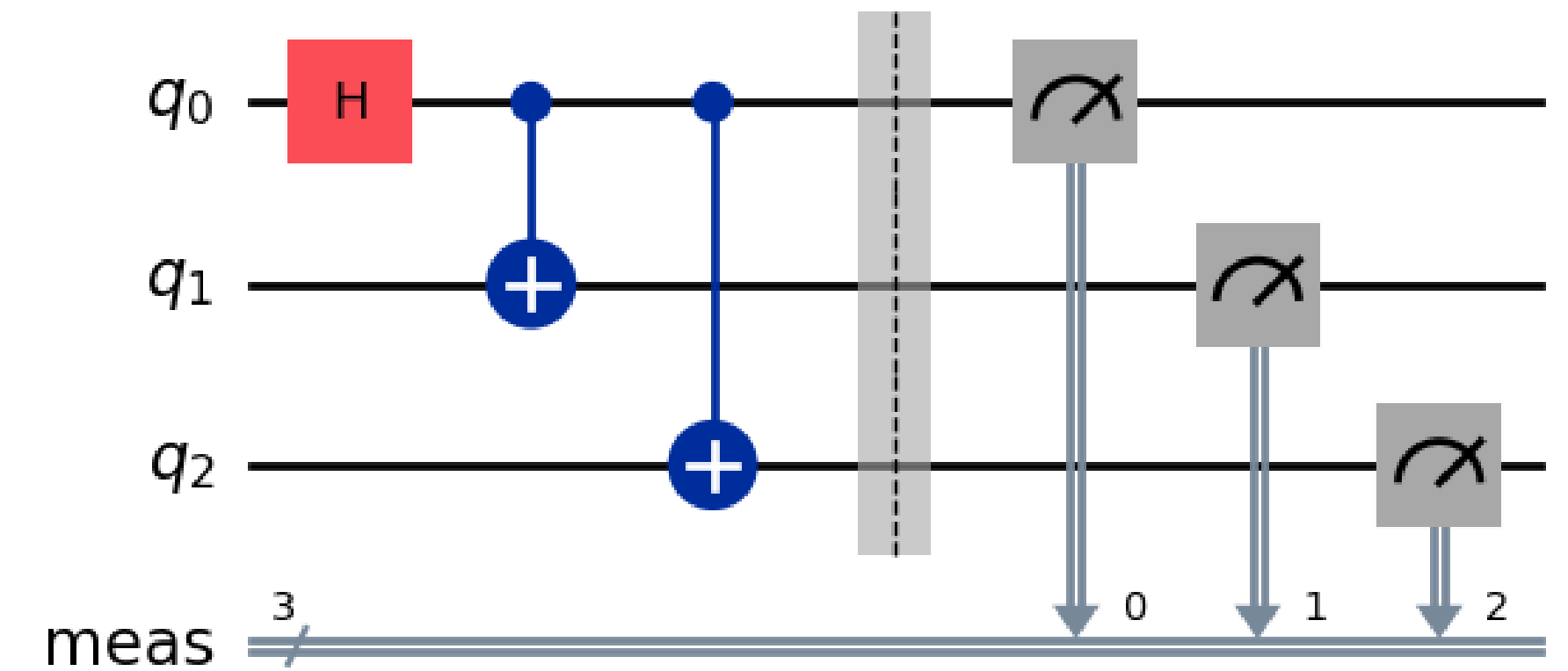
Quantum bits (qubits) and quantum circuits



A **quantum bit** or qubit is a controllable quantum object that is the unit of information



A **quantum** device is made up of physical qubits, fixed in place on the chip



A **quantum** circuit is a set of quantum gate operations on qubits and is the unit of computation

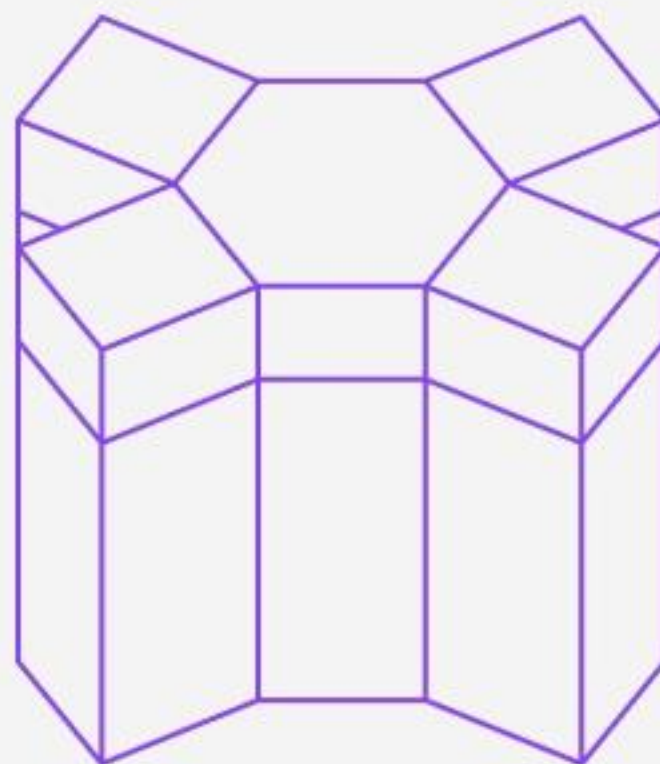
Quantum must be
performant

Quantum must be
easy-to-use



Qiskit

+



IBM QPUs

=



Utility Scale Work

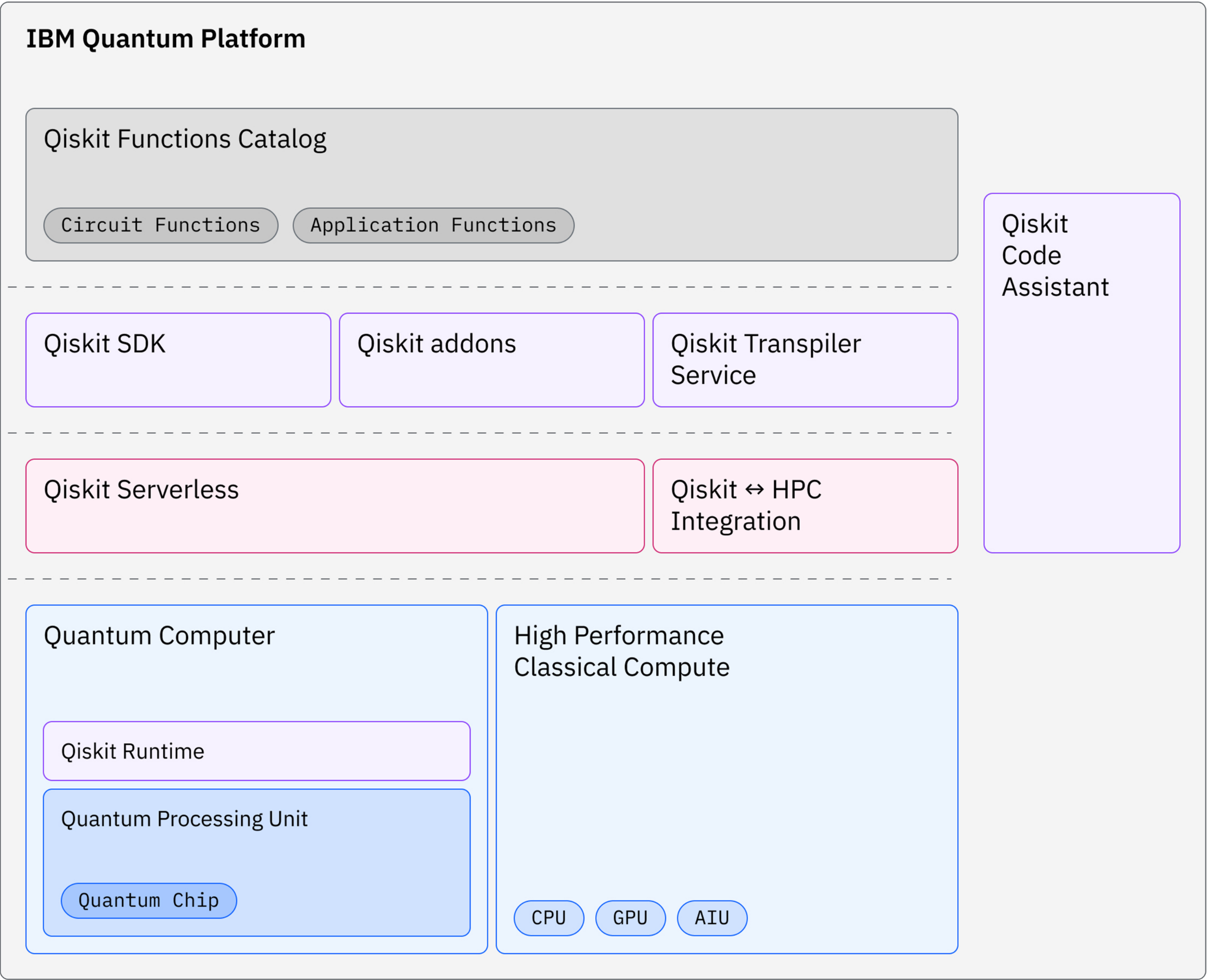


Abstraction
Services

Dev Tools

Heterogeneous
Orchestration Layer

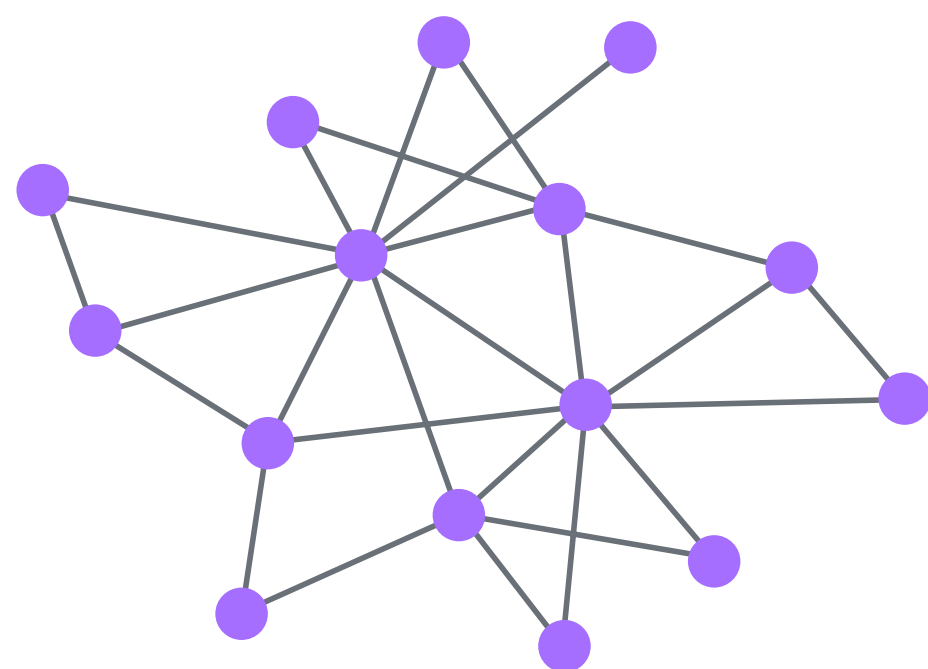
Compute



The Workflow: Qiskit patterns

Step 1

Map classical inputs to a quantum problem



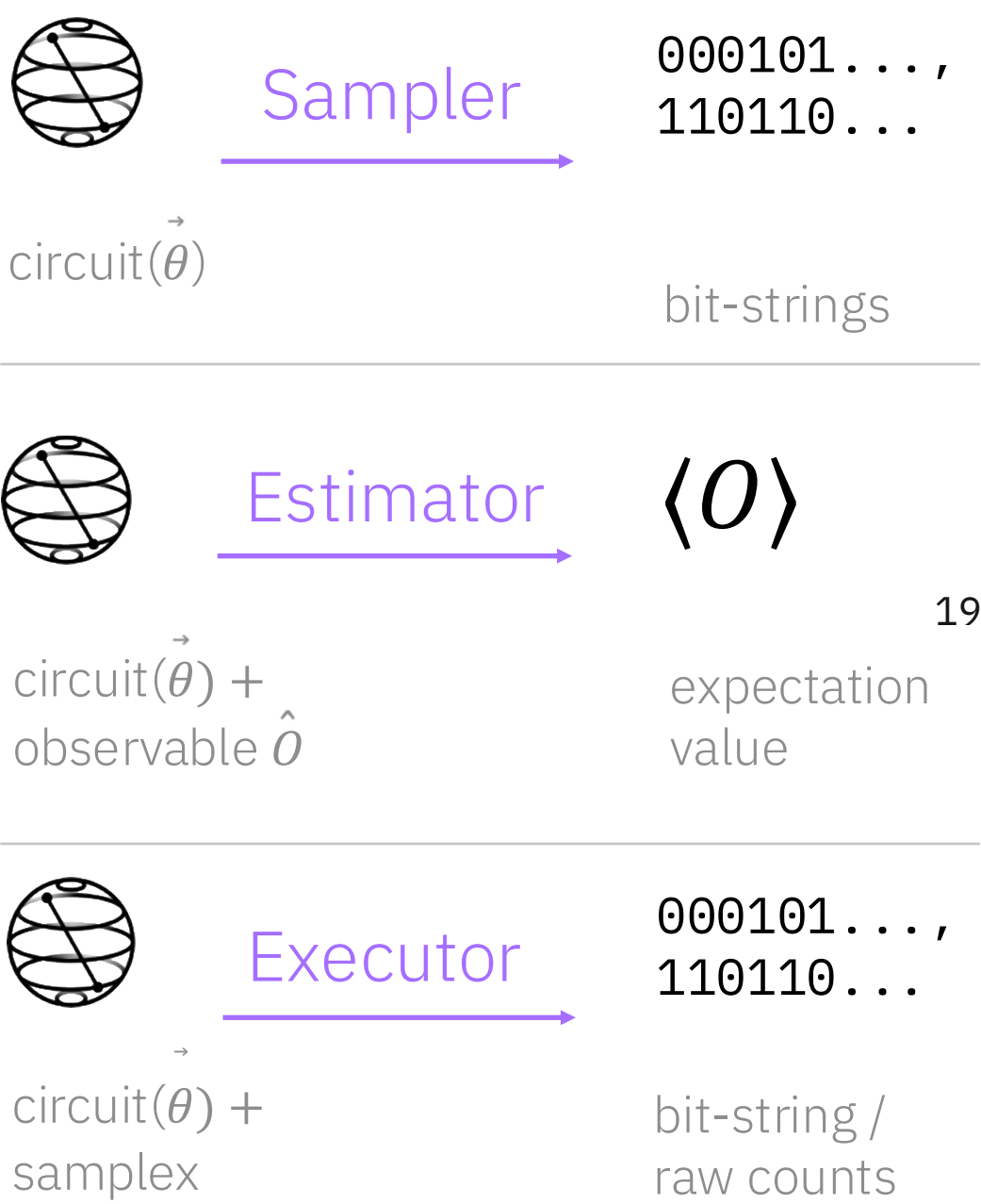
Step 2

Optimize problem for quantum execution.

```
PassManager([UnitarySynthesis(),
              BasisTranslator(),
              EnlargeWithAncilla(),
              AISwap(),
              Collect1qRuns(),
              Optimize1qGates(),
              Collect2qBlocks(),
              ConsolidateBlocks()])
```

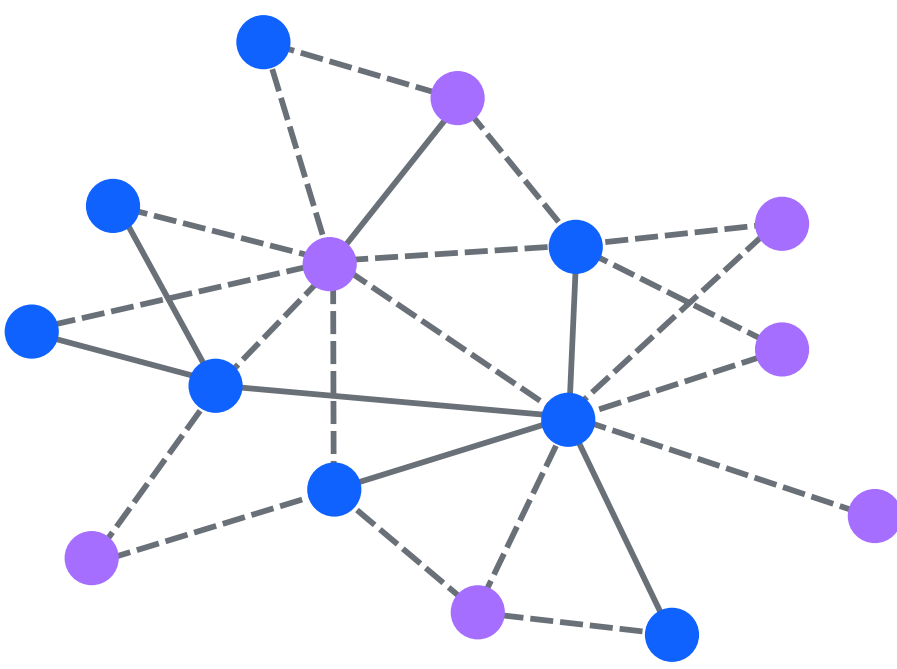
Step 3

Execute using Qiskit Runtime Primitives.



Step 4

Post-process, return result in classical format.



IBM Quantum Platform

To bring useful quantum computing to the world, quantum computing must be easy to use.

To support this mission, we’ve gathered all IBM Quantum resources into one easy-to-access location.

Our goal is to build a mature and stable quantum programming experience.

IBM Quantum Platform

API Token

.....

Explore announcements from the inaugural IBM Quantum Developer Conference. [Watch the event replay](#)

Open Plan

[View details](#) | [Upgrade](#)

Up to 10 minutes/month

Monthly usage

Used

0ms

Remaining

10m

Recent workloads

0

Pending

5

Finished workloads

ID

Status

Completed

Mode

Compute resource

Usage

ibmq_16mq_v2

Completed

28 Nov 2023

ibmq_16mq_v2

ibmq_16mq_v2

100%

ibmq_16mq_v2

Completed

28 Nov 2023

ibmq_16mq_v2

ibmq_16mq_v2

100%

ibmq_16mq_v2

Completed

28 Nov 2023

ibmq_16mq_v2

ibmq_16mq_v2

100%

ibmq_16mq_v2

Completed

28 Nov 2023

ibmq_16mq_v2

ibmq_16mq_v2

100%

ibmq_16mq_v2

Completed

28 Nov 2023

ibmq_16mq_v2

ibmq_16mq_v2

100%

Instance QPUs

→

3



All QPUs

→

13

Documentation

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Q

Hello World

Create a simple quantum program and run it on a quantum processing...

Qiskit Runtime

Introduction to primitives

Learning

[Open app](#)

Featured course

Quantum Computing in Practice

Catalog

Explore all courses and tutorials

IBM Quantum Composer

Graphically build circuits



What's new

→

Service alert

Brief maintenance event affecting ibmq_strasbourg and ibmq_brussels

3 days ago • [Read more](#)

Blog

IBM Quantum delivers on performance challenge

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Product update

Fractional gates to become part of the Heron instruction set architecture

18 days ago • [Read more](#)

Product update

What's new in the docs?

About 1 month ago • [Read more](#)

Blog

A closer look at Qiskit Code Assistant

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Product update

Upcoming sunset of backend.run

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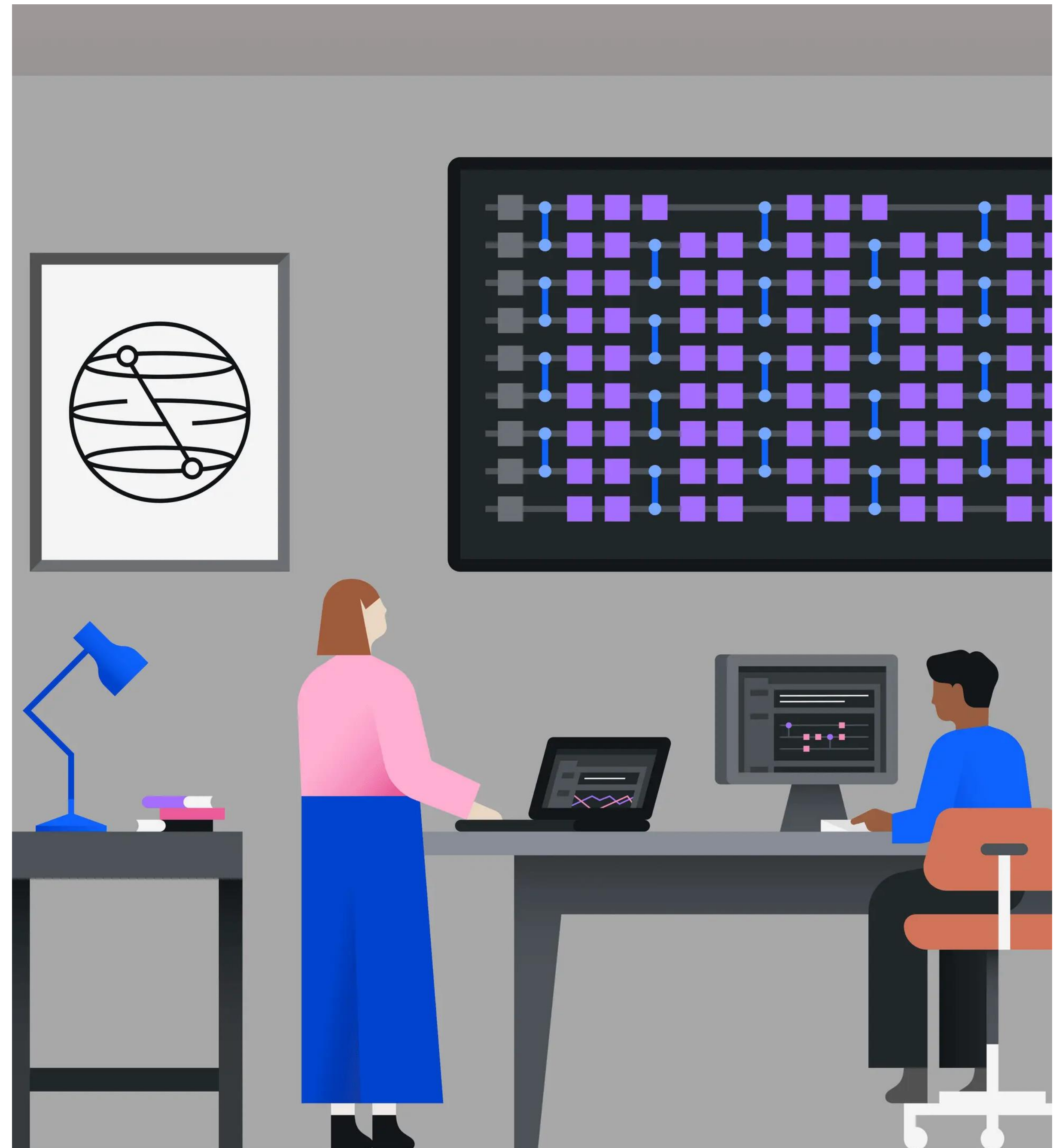
Begin exploring quantum today with the Open Plan

Get up to 10 minutes of runtime per month free.

- ❑ Access to IBM Quantum Compute service
- ❑ Access to 100+ qubit quantum computers

Create an account on IBM Quantum Platform and start using 100+ qubit IBM quantum computers today!

PROMO: get 180 minutes IBM QPU runtime free when you use at least 20 minutes over a 12-month period.

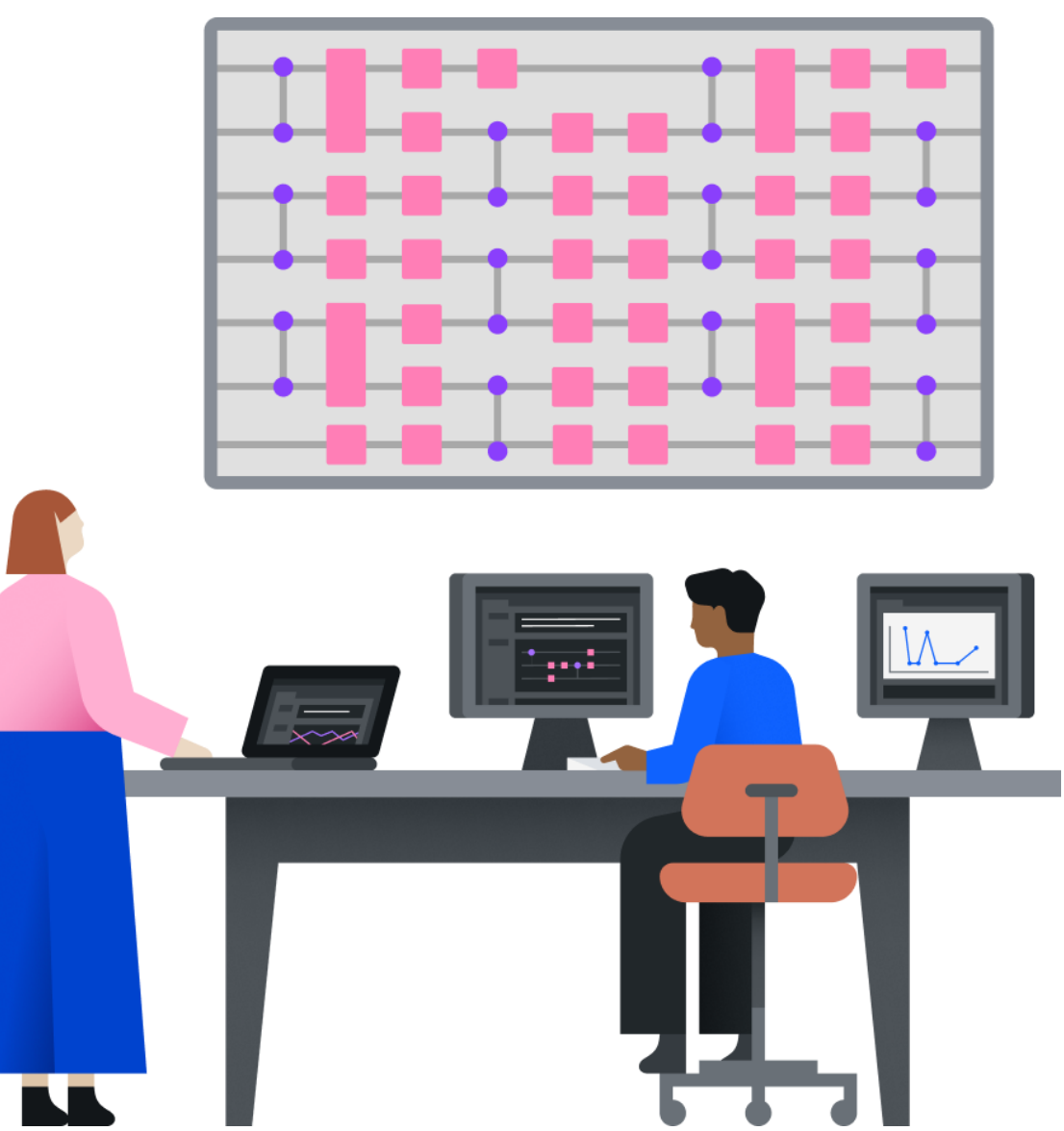


Next steps!



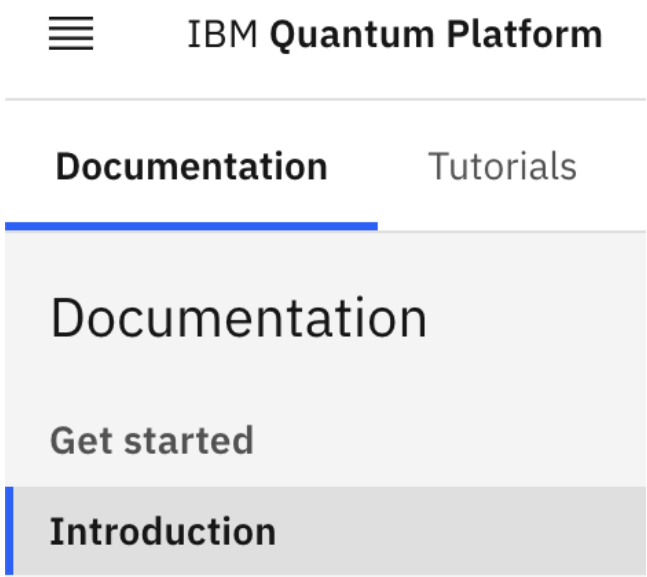
IBM Quantum Learning Platform

An online platform for learning the basics of quantum computing, and how to use IBM Quantum services and systems to solve real-world problems.



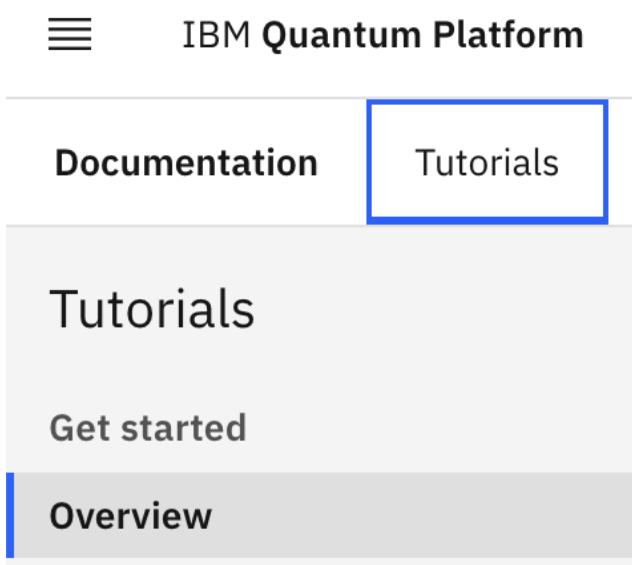
Documentation

The starting point whether you want to simply get started or you are an expert looking for a specific answer.



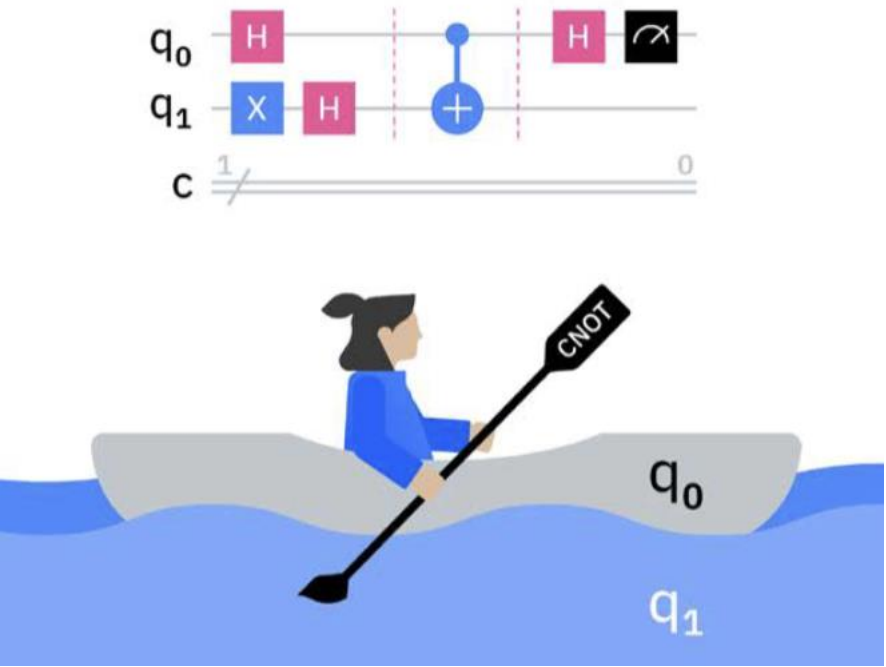
Tutorials

Improve your coding skills and learn how to apply Qiskit to common use cases.



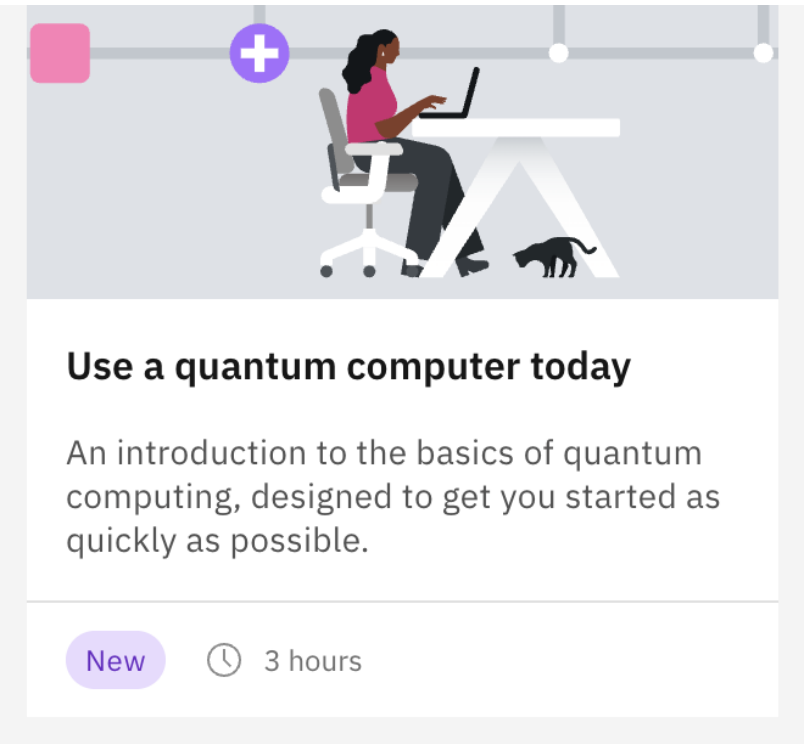
Modules

Designed to help incorporate quantum computing into traditional STEM classrooms.

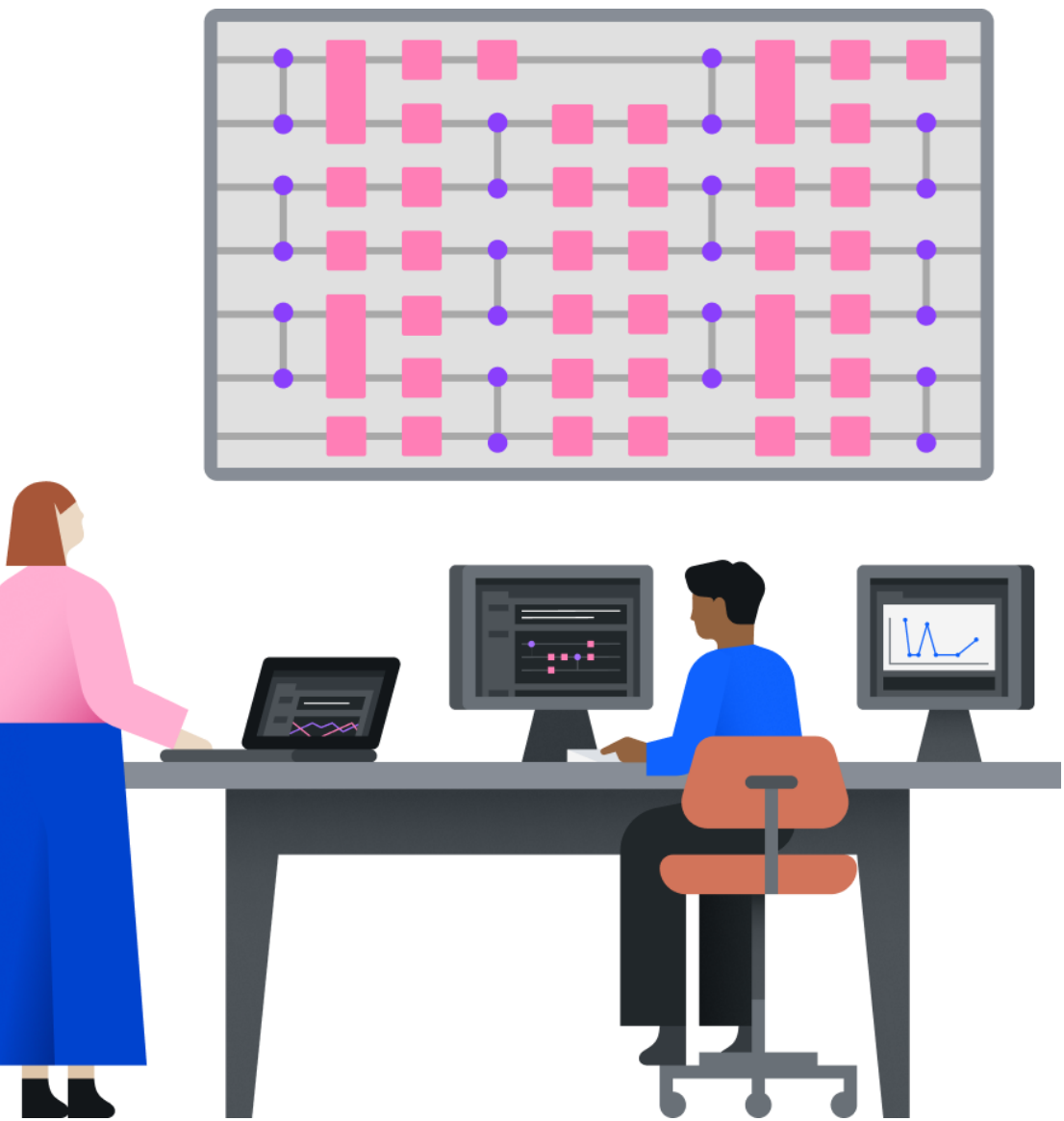


Courses

Start learning and applying quantum computing with Qiskit through our library of 10+ courses from leading experts.

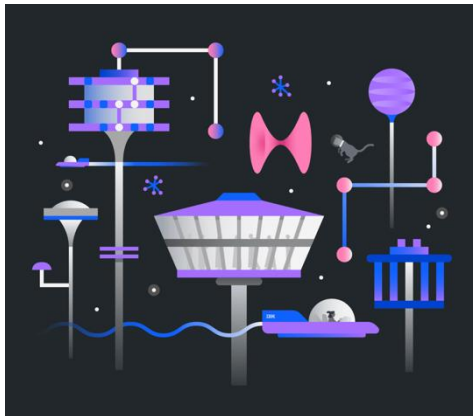


IBM Quantum Community



Qiskit Fall Fest

Student-led quantum computing events on campus
↳ Crafted by student leaders, supported by IBM



Qiskit Advocates

A global program that provides support to individuals who actively contribute to the Qiskit community.



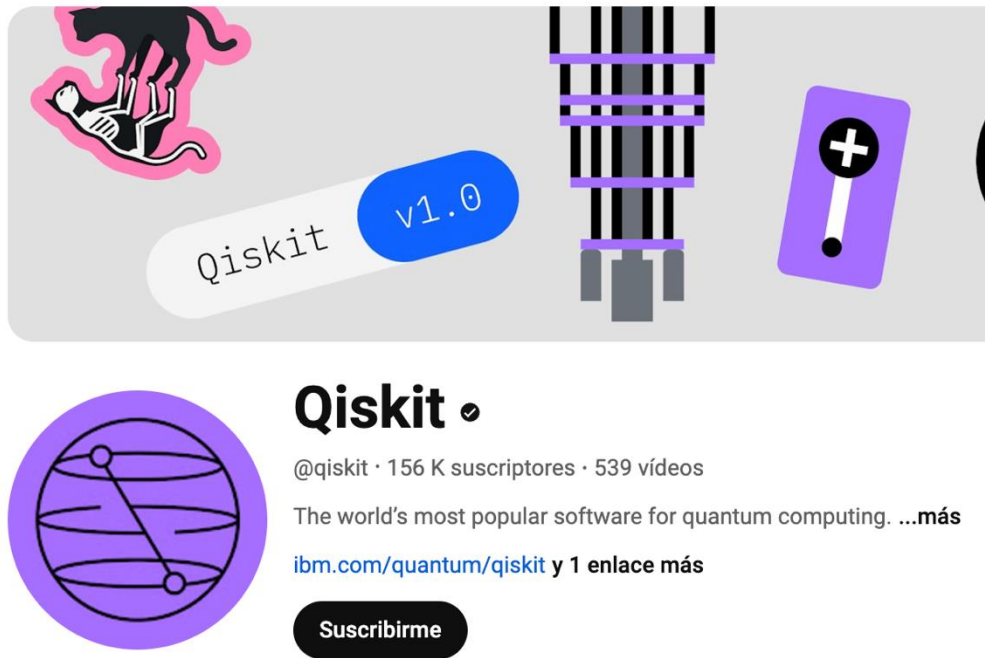
Qiskit Global Summer School

An annual event featuring online lectures delivered by various IBM Quantum experts, as well as live Q&A sessions.



Qiskit YouTube Channel

The Qiskit YouTube channel hosts hundreds of useful videos on quantum computing.



Qiskit Developer Certification

The world's first ever developer certification for programming a quantum computer, setting the benchmark for quantum developer skills.



IBM Quantum Developer Conference

A conference that brings together developers from all over the world to preview the hardware and software that will power the first demonstrations of quantum advantage on IBM quantum computers.

Qiskit Youtube

Engaging lectures, tips & tricks, tutorials, community updates and access to exclusive Qiskit content.

Notable Playlists:

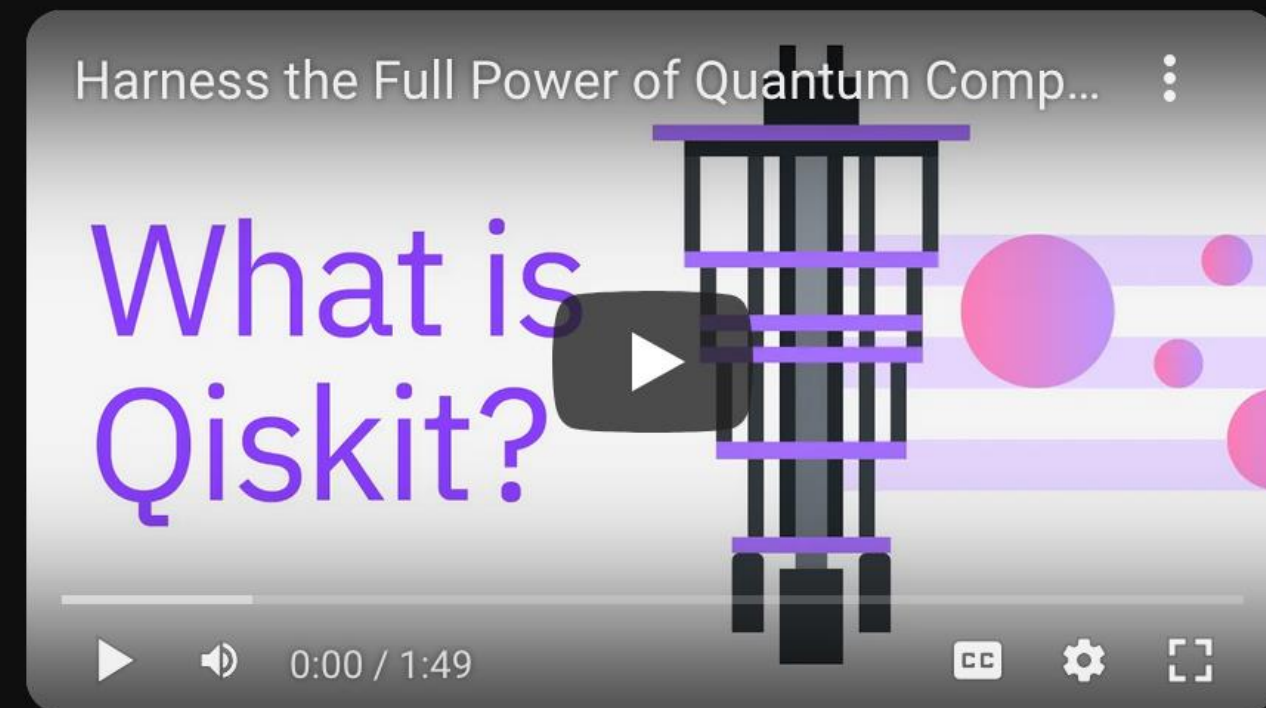
[Qiskit in the Classroom](#)

[Quantum Computing in Practice](#)

[Quantum Crosstalk](#)

[Qiskit Quantum Seminars](#)

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Harness the Full Power of Quantum Computers

5,061 views • 2 weeks ago

Qiskit is a high-performance software stack built to help developers and researchers harness the full power of quantum computers at the utility scale and beyond.

Get started with Qiskit today

<https://learning.quantum.ibm.com/lear...>

Quantum Computing in Practice ▶ Play all

Many videos on quantum computing focus on their theoretical concepts or future potential, but in this series, Dr. Olivia Lanes will show you how to use a quantum computer that exists right now to...



Quantum Computing in Practice Series Trailer



11K views • 10 months ago



Your Guide to 100+ Qubits: Quantum Computing in...



77K views • 10 months ago



Your Guide to Quantum Circuits and Qiskit: Quantu...



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Your Guide to Utility Scale QAOA: Quantum Computing...



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Which Problems Are Quantum Computers Good...



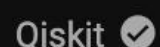
12K views • 4 months ago

Quantum Crosstalk ▶ Play all

The short form show where we sit down with influential members of IBM's quantum team and discuss the latest happenings, events, and product releases.



Qiskit Performance with Dr. Blake Johnson | Quantum...



2.1K views • 5 months ago



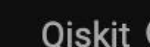
What It's Like to Intern at IBM Quantum | Quantum...



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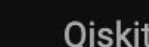
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What's Coming to Quantum in 2024: Quantum Crosstalk...



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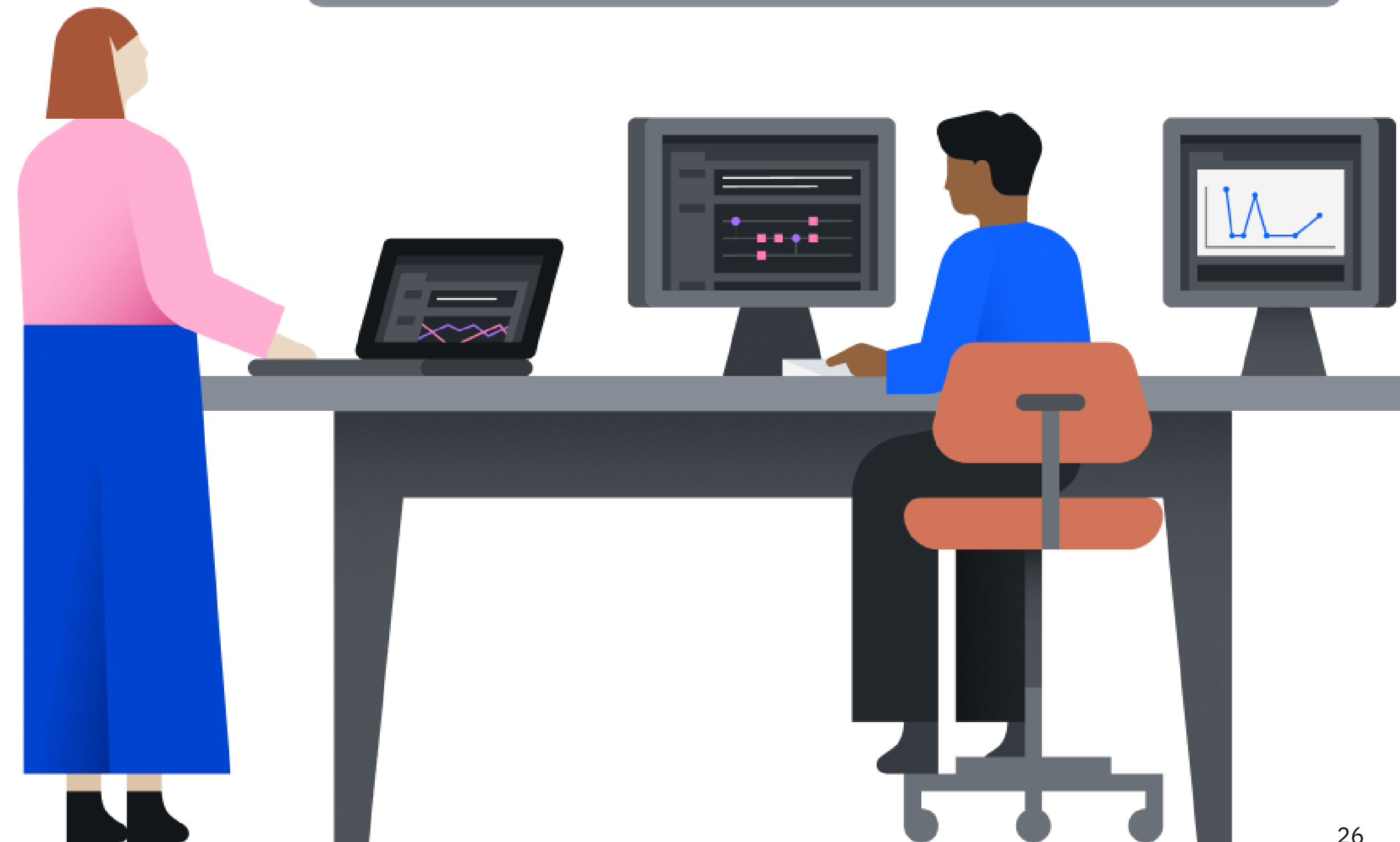
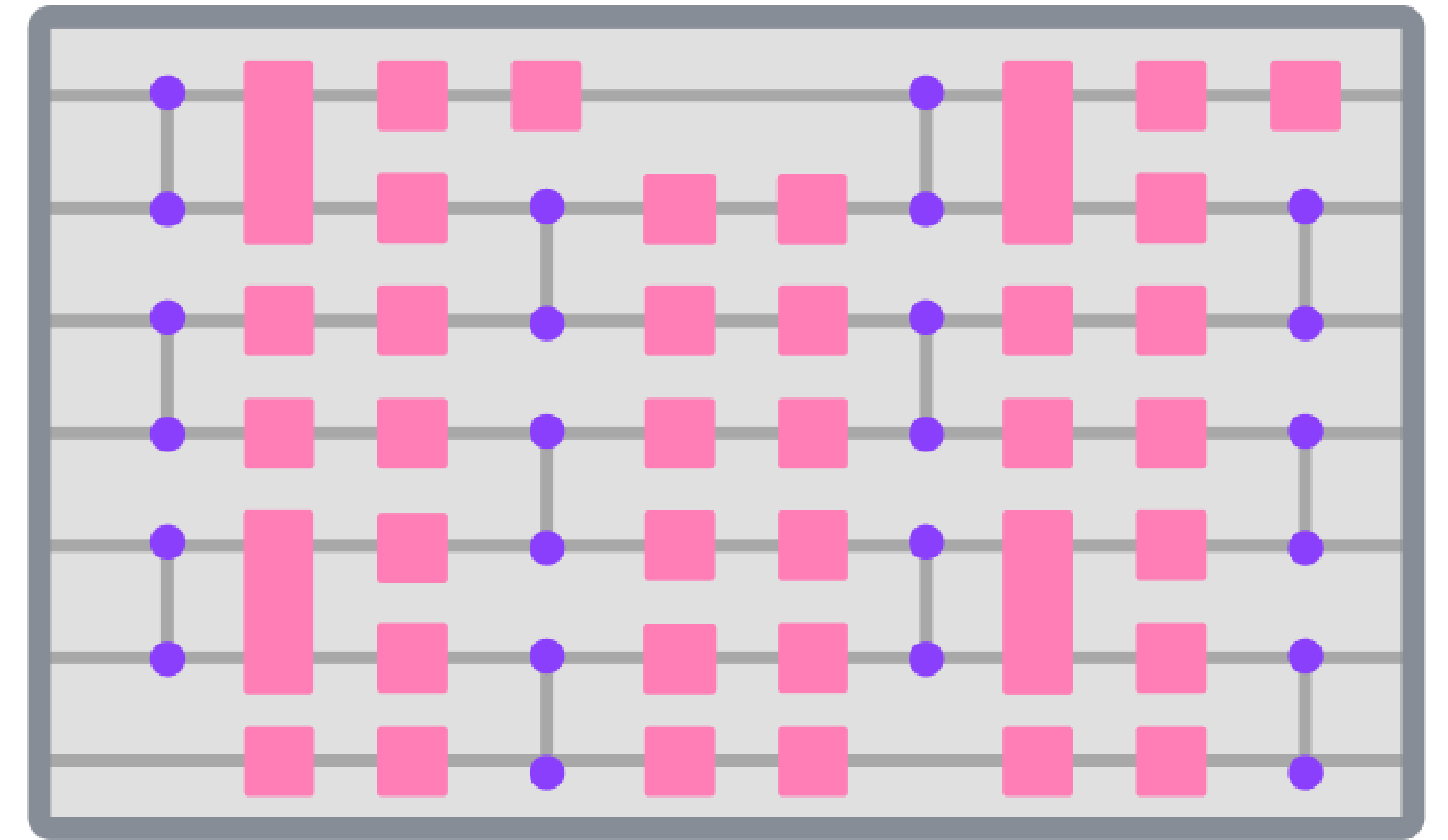
Qiskit Global Summer School

Our largest educational event every year. Two weeks of lectures, labs, Q&As, and community engagement.

Rooted in the IBM Quantum Learning courses, with new content each year.

Live event featuring badges for completion, extra assignments for partners, and more.

After the event is over, the lecture material becomes available on YouTube.



Qiskit Fall Fest



Student-led quantum computing events on campus

↳ Crafted by student leaders, supported by IBM

Working with the student leaders of today to shape the industry leaders of tomorrow

Organic growth of local communities

Since 2021:

500+ 60k+

Events

Attendees

IBM Quantum Summer Internship Program

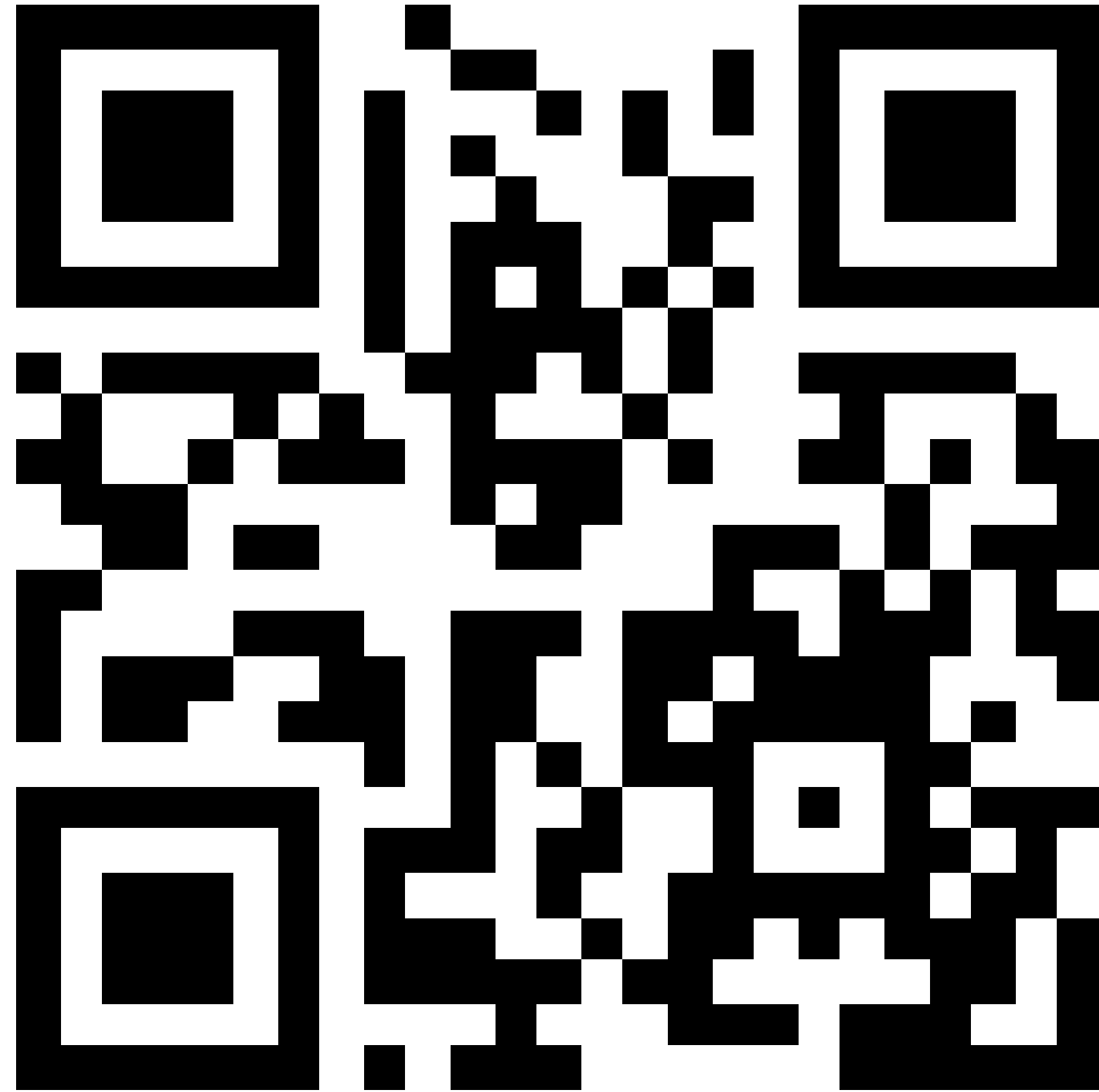
Work across the entire IBM Quantum team, from programming to materials research, on real projects that matter.

Applications open in the fall and continue through winter for different roles globally.



IBM Quantum interns, class of 2023

Scan here to learn all things Qiskit Fall Fest and resources!



IBM Quantum